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Institution

Month, Year

To my family.

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I would like to thank my supervisor, **Prof. Supervisor Name**, for the guidance and support throughout this work.

Resumo

Este trabalho aborda o problema de ...

Palavras-chave: palavra-chave 1, palavra-chave 2, palavra-chave 3.

Abstract

This thesis addresses the problem of . . .

Keywords: keyword 1, keyword 2, keyword 3.

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CHAPTER 1

Introduction

This chapter introduces the addressed problem, motivates the work, states the research questions, summarises the contributions, and outlines the remainder of the document.

1.1. Motivation

1.2. Problem Statement

1.3. Research Questions

This work addresses the following research questions:

RQ1.

RQ2.

RQ3.

1.4. Objectives

The main objectives are:

-
-

1.5. Contributions

The main contributions are:

-
-

1.6. Document Structure

The remainder of this document is organised as follows. Chapter 2 reviews the relevant literature and prior work. Chapter 3 describes the proposed approach. Chapter 4 presents and discusses the experimental results. Chapter 5 summarises the work and outlines directions for future work.

CHAPTER 2

Literature Review

This chapter surveys the existing literature relevant to left-ventricle reconstruction from short-axis cardiac MRI. We focus on three connected themes: the problem of reconstructing the left ventricle from short-axis magnetic resonance images, the use of statistical shape models to regularize cardiac geometry, and point-based implicit representations that reconstruct smooth surfaces from sparse contour observations.

2.1. Background

Short-axis cardiac magnetic resonance imaging is widely used because it offers clear anatomical contrast and standardized coverage of the left ventricle, but the imaging geometry remains fundamentally slice-based rather than volumetric (Cerqueira et al., 2002; Petitjean & Dacher, 2011). This creates a reconstruction problem: the left ventricle must be inferred in three dimensions from a stack of two-dimensional contours or partial surfaces, and the quality of the result depends on how well the method can interpolate between slices while preserving global shape and local wall geometry (Heimann & Meinzer, 2009).

The literature on cardiac shape analysis makes a consistent distinction between methods that learn a compact representation of shape and methods that directly operate on image or mesh topology. Compact representations are attractive because they reduce the effect of noise and missing data, but they also impose a limited deformation space. Direct methods are more expressive, but they tend to require more data and stronger inductive bias to remain anatomically plausible (Bai et al., 2015; Heimann & Meinzer, 2009; Suinesiaputra et al., 2014).

Statistical shape models are therefore a natural bridge between the imaging problem and the downstream geometric analysis. In the cardiac setting, they are used to encode variation across subjects, to regularize segmentation, and to derive measurements such as ventricular volume or wall thickness (Heimann & Meinzer, 2009; Suinesiaputra et al., 2018).

2.2. Related Work

The literature falls into three families. Early work on left-ventricle reconstruction focused on interpolation and parametric surface fitting, later work emphasized statistical shape models, and more recent approaches adopt point-set encoders and implicit surface representations to model geometry at continuous resolution (Heimann & Meinzer, 2009; Mescheder et al., 2019; Park et al., 2019; Petitjean & Dacher, 2011; Qi et al., 2017; Tancik et al., 2020).

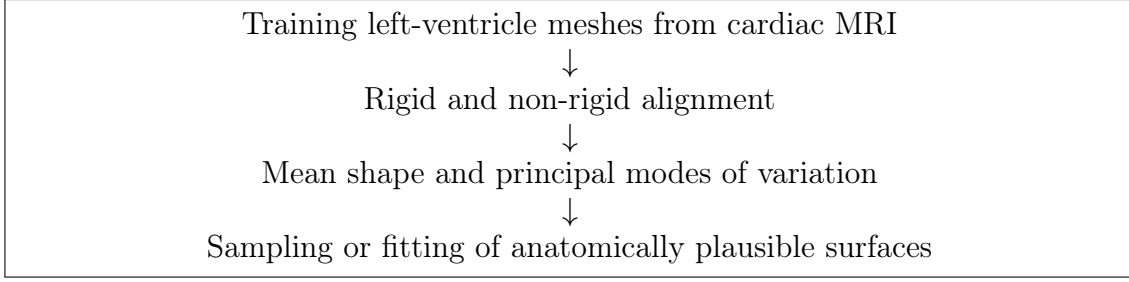


FIGURE 2.1. Schematic of a statistical shape model for the left ventricle. The goal is to compress a population of aligned meshes into a low-dimensional shape space that can later be sampled or fitted to new data.

TABLE 2.1. Representative method families for left-ventricle reconstruction and wall-thickness analysis.

Family	Core idea	Strengths	Limitations
Interpolation and parametric fitting	Infer the 3D surface from stacked slices or from a low-dimensional analytic shape prior (Heimann & Meinzer, 2009; Petitjean & Dacher, 2011).	Simple, fast, and easy to interpret.	Sensitive to sparse coverage and may oversmooth local morphology.
Statistical shape models	Learn a population shape space and fit it to new cases (Bai et al., 2015; Cootes et al., 1995; Heimann & Meinzer, 2009).	Strong anatomical prior and robust to missing information.	Limited by the training population and by the linearity of the latent space.
Point-based and implicit representation learning	Encode unordered point clouds with a permutation-invariant encoder and recover the surface as a continuous level set (Mescheder et al., 2019; Park et al., 2019; Qi et al., 2017; Tancik et al., 2020).	Handles sparse or irregular samples and supports sub-voxel surface detail.	Depends on a good sampling strategy and on an implicit decoder that is stable to train.

2.2.1. Interpolation and classical shape fitting

Classical reconstruction methods usually begin with a set of segmented SAX slices and then interpolate between them or fit a parametric model to the observed contours. Reviews of cardiac segmentation note that such methods are attractive because they are computationally simple, but their performance is limited by the absence of direct information between slices, especially near the apex and base of the ventricle (Heimann & Meinzer, 2009; Petitjean & Dacher, 2011). In practice, these methods tend to encode geometric assumptions explicitly, so their accuracy depends on how closely those assumptions match the anatomy of the patient being reconstructed (Heimann & Meinzer, 2009).

The main advantage of this family is interpretability. When a reconstruction is driven by interpolation or by a low-order geometric prior, failures are easier to inspect and debug than in a purely learned pipeline. The main drawback is that the model does not learn a rich data-driven representation of cardiac variation, so it may underfit subjects whose ventricle geometry departs from the training assumptions (Bai et al., 2015; Heimann & Meinzer, 2009).

2.2.2. Statistical shape models

Statistical shape models provide a compact representation of anatomical variation by aligning a training set of meshes or contours and then learning the dominant modes of variation, typically through principal component analysis (Cootes et al., 1995; Heimann & Meinzer, 2009). For the left ventricle, this framework is especially useful because it can absorb missing slices, suppress segmentation noise, and constrain the reconstructed surface to remain within a plausible cardiac manifold (Suinesiaputra et al., 2014).

Cardiac-specific studies have shown that statistical atlases and shape models can be built from large cohorts and used to study normal variation, disease-related remodelling, and motion patterns (Bai et al., 2015; Khalafvand et al., 2018; Medrano-Gracia et al., 2014). In particular, the left ventricle does not vary only by size; it also varies by eccentricity, curvature, and regional deformation, which makes the choice of shape basis important for later measurements (Medrano-Gracia et al., 2014; Suinesiaputra et al., 2018).

The main limitation of a linear statistical model is that it compresses the population into a low-dimensional subspace. This is useful when the goal is regularization, but it can become restrictive when the anatomy is complex or when the input slices are sparse (Bai et al., 2015; Heimann & Meinzer, 2009).

2.2.3. Point-based and implicit representation learning

PointNet introduced a simple but effective idea for unordered geometric data: an encoder can consume a point set directly if the architecture is permutation invariant and aggregates local features into a global latent code (Qi et al., 2017). This matches the implemented notebook pipeline, which starts from contour-derived point samples and learns a latent shape description before decoding geometry.

The decoder side of the model belongs to the implicit surface family. DeepSDF represents a shape as the zero level set of a learned signed-distance function, which makes completion from partial input and shape interpolation natural (Park et al., 2019). Occupancy Networks use a closely related idea, but predict occupancy instead of signed distance; the common point is that the surface is reconstructed as a continuous decision boundary rather than as a fixed polygonal template (Mescheder et al., 2019).

The inspected training notebook clarifies the implemented practical direction. The model is explicitly phase-conditioned and predicts paired endocardial and epicardial signed-distance fields with a coupled parameterization that preserves positive wall thickness by construction. In the same implementation, Fourier feature mappings and SDF-oriented

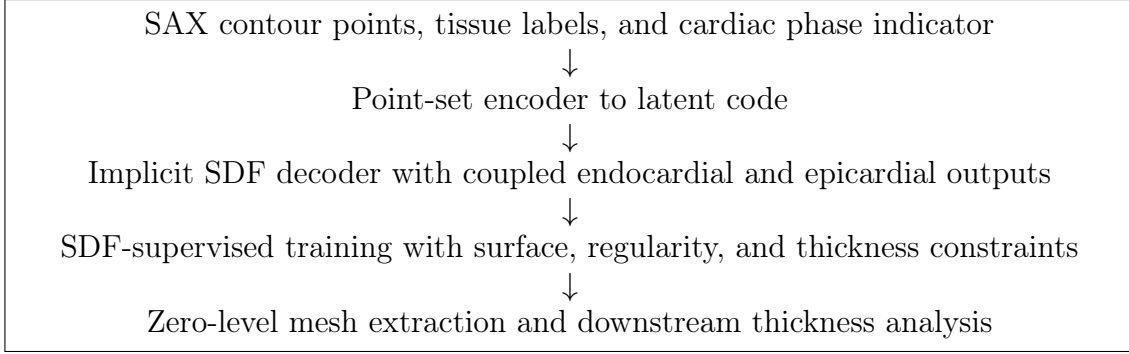


FIGURE 2.2. Placeholder schematic of the phase-conditioned implicit SDF reconstruction pipeline aligned with the training notebook implementation.

losses concentrate supervision on surface fidelity, gradient regularity, and off-surface consistency, while mesh extraction is performed from the zero level set with marching cubes. This design is consistent with implicit SDF reconstruction literature (Mescheder et al., 2019; Park et al., 2019; Qi et al., 2017; Tancik et al., 2020).

Fourier feature mappings are relevant because they mitigate the spectral bias of plain multilayer perceptrons and help them fit high-frequency geometric detail (Tancik et al., 2020). In practice, this matters when the model must preserve thin walls, sharp curvature changes, or small local variations in the reconstructed ventricle surface.

In the implemented pipeline, the key point is the combination of a PointNet-style encoder with an implicit SDF decoder. That combination is more consistent with the actual notebooks because it learns a compact latent code from unordered samples and decodes a continuous surface that can be interrogated at arbitrary resolution (Park et al., 2019; Qi et al., 2017; Tancik et al., 2020).

2.2.4. Wall-thickness analysis

Wall thickness is a clinically meaningful quantity because it reflects cardiac remodelling and myocardial health, and it is typically measured as the distance between endocardial and epicardial surfaces or through equivalent geometric constructs (Cerqueira et al., 2002; Grossman et al., 1974). Recent MRI studies continue to treat wall thickness as a sensitive marker of myocardial structure and tissue properties (Huehnhaagen et al., 2017).

The literature therefore links the reconstruction problem to a downstream measurement problem. A method that reconstructs the left ventricle more faithfully will usually yield more stable thickness estimates, whereas a method that over-smooths the surface can attenuate local thickness variation (Grossman et al., 1974; Huehnhaagen et al., 2017). This is why wall-thickness estimation cannot be treated as a detached post-processing step: it depends on the geometric fidelity of the reconstructed surface itself (Cerqueira et al., 2002; Medrano-Gracia et al., 2014).

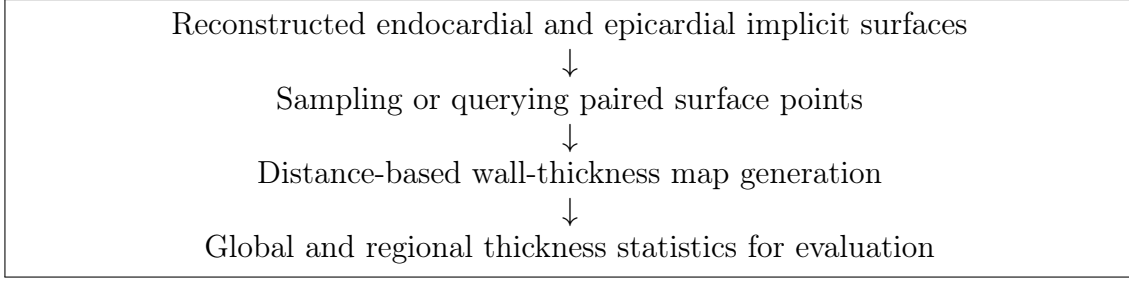


FIGURE 2.3. Placeholder diagram for the wall-thickness analysis workflow from paired ventricular surfaces.

2.3. Summary and Research Gap

The reviewed literature shows a clear progression. Classical interpolation and parametric fitting methods are simple but limited. Statistical shape models add a strong anatomical prior and are well suited to cardiac anatomy, but they are restricted by the learned latent space. Point-based implicit methods offer continuous surface modelling from sparse observations, yet they require careful sampling and regularization to remain anatomically stable (Bai et al., 2015; Heimann & Meinzer, 2009; Park et al., 2019; Qi et al., 2017; Tancik et al., 2020).

The identified gap is the combination of these strengths in a single pipeline. Existing studies usually emphasize either population-level shape priors or generic implicit reconstruction, but less often a unified workflow that starts from sparse SAX contours, conditions reconstruction on cardiac phase, and preserves downstream thickness consistency as part of model design (Grossman et al., 1974; Huelnhagen et al., 2017; Park et al., 2019; Qi et al., 2017; Suinesiaputra et al., 2018). The remaining chapters describe a pipeline that combines statistical shape modelling, PointNet-style encoding, phase-conditioned implicit SDF decoding, and wall-thickness-oriented evaluation.

CHAPTER 3

Methodology

This chapter describes the proposed approach to address the research questions stated in Section 1.3.

3.1. Overview

3.2. System Architecture

3.3. Components

3.3.1. Component A

3.3.2. Component B

3.4. Implementation Details

CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

4.1.1. Datasets

4.1.2. Metrics

4.1.3. Baselines

4.2. Results

4.3. Discussion

4.4. Threats to Validity

CHAPTER 5

Conclusions

This chapter summarises the contributions, revisits the research questions, and outlines directions for future work.

5.1. Summary

5.2. Revisiting the Research Questions

5.3. Contributions

5.4. Limitations

5.5. Future Work

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APPENDIX A

Appendix Title

A.1. Section in the Appendix